

PART 1: 📄 CROSS-QUESTIONS PDF (Light → Higher Level)

Cross-Question Bank — PS 26034

Level 1 (Easy) → Level 5 (Expert/Killer) — With Model Answers

□ LEVEL 1 — BASIC UNDERSTANDING (Warm-up Questions)

Q1. Problem kya hai, simple words me?

India ke packaged products par mandatory declarations (MRP, net quantity, manufacturer details, dates) hote hain jo Legal Metrology Rules ke under required hain. Crores of products, lekin limited manual inspectors — violations jaise chhota font, galat ya missing declaration, dual-MRP pakda nahi jaata. Hum AI + OCR se automated checking banate hain.

Q2. Tumhara solution ek line me kya hai?

Photo lo → hamara system label padhta hai (OCR), saare mandatory declarations extract karta hai, Indian Legal Metrology Rules ke against validate karta hai, aur enforcement-grade PDF report deta hai — seconds me.

Q3. Kaun use karega ye system?

Primary: Legal Metrology inspectors aur department officials. Secondary: manufacturers self-audit ke liye, e-commerce platforms bulk checking ke liye.

Q4. Ye online/offline dono kaam karta hai?

Haan. Offline mode me on-device quick preview milta hai; internet aane par full validation server pe hoti hai.

Q5. Cost kitna lagega government ko?

Software-only cost — koi extra hardware nahi, sirf existing smartphones. Manual system ki manpower cost se zyada costly nahi, lekin coverage 100x badhata hai.

□ LEVEL 2 — SCOPE & DOMAIN CLARITY

Q6. Kya ye packet ke andar ki quality/ingredients test karega?

Nahi — wo FSSAI/lab ka domain hai. Hum label-par-kya-lika-hai check karte hain (presence, format, font, placement). Ingredients PANEL ka existence aur formatting hamara scope hai; content ki sachchai lab testing hai.

Q7. Net quantity actually kam bhara ho to?

Lab-scale verification hum scope nahi. But Phase-3 me inspector spot-weigh data entry + systematic short-fill pattern detection possible hai.

Q8. Konse rules cover karte ho?

LM (PC) Rules 2011 — Rule 4 (name/address), 5 (net quantity), 6 (MRP/care details), 7 (month), 9(6) (font size), 10 (e-commerce), 11 (imports). Rules DB me versioned hain — amendments updateable.

Q9. Rules change hote rahenge to system outdated ho jayega?

Rules code me hardcoded NAHI hain — DB entity hain with versioning aur effective_from date. Government amendment aaye to admin UI se update; historical scans purane rules se evaluate rehte hain.

Q10. State-wise rules alag ho to?

Central Act hai but state-level variations exist karte hain. Rule engine scan ke GPS/state ke hisaab se applicable rule versions load karta hai.

□ LEVEL 3 — TECHNICAL DEPTH

Q11. Font size 1mm camera se accurately measure kaise loge? Perspective?

Reference-object calibration: barcode module width (known 0.33mm EAN standard) ya coin edge se px-to-mm scale derive karte hain. Realistic error $\pm 10\text{--}15\%$ — isliye borderline cases "MANUAL VERIFY" flag hote hain, automatic fail nahi. Controlled-sheets se accuracy validated karenge.

Q12. OCR curved bottles/shiny wrappers pe fail ho jayega?

Quality gate pehle reject karta hai jo process na ho sake. Curved text ke liye CRAFT-style detection + unwarping preprocessing. Per-field confidence score — low confidence human-in-loop, silent pass nahi.

Q13. Training data kahan se aayega? Ready-made dataset hai?

India-specific dataset exist nahi karta — ye hi hamara moat hai. Synthetic label generation + augmentation, public OCR datasets (COCO-text etc.) pretraining ke liye, aur presence/format checks annotation-free hain. Production me inspector corrections se continuously improve hota hai.

Q14. Scale kya hoga? Daily scans kitne handle honge?

10TB/month). Hackathon MVP single server pe bhi runnable.

Q15. Duplicate product kaise detect karte ho — same product different photo angle?

Barcode primary identifier. Nahi mile to composite fingerprint: perceptual hash + brand-logo embedding similarity + extracted text fuzzy match.

Q16. Frontend-backend kaise communicate karte hain?

REST API (JWT-secured) + WebSocket for live job status. Upload pe async 202+job_id — polling nahi, push notification.

Q17. Database schema me sabse important design decision kya hai?

(a) Rules as separate versioned entity — amendments ke liye. (b) Violation linked to BOTH scan AND specific extracted field — legal evidence traceability. (c) Immutable audit log.

● LEVEL 4 — LEGAL, ETHICS, ACCOUNTABILITY

Q18. AI-generated notice court me valid hogi?

Hum notice ISSUE nahi k, sirf DRAFT karte hain — approval human Controller ka hai. Evidence-backed digitally-signed report tamper-proof hai — manual handwritten memos se zyada defensible. Golden phrase: *"AI assists, humans decide."*

Q19. False positive hua to? Manufacturer ka loss kaun bharega?

Three-tier output: CONFIRMED / SUSPECTED / PASS. Sirf CONFIRMED workflow me jaata hai. Raw evidence attached, dispute portal available, false-positive rate tracked as KPI (<10% target).

Q20. Inspector khud corrupt ho — fake photo daal de?

Gallery upload blocked (camera-I only), EXIF+GPS cryptographically signed at capture, audit-log immutable, statistical anomaly detection on inspector patterns. Corruption-proof nahi, manipulation COSTLY aur DETECTABLE banata hai.

Q21. Vendor ka private data DPDP concerns?

Personal consumer data collect hi nahi hota. Vendors sirf apna data dekhte hain (RBAC). Enforcement-derived data public matter. Encryption at rest/in transit, DPDP 2023-aligned.

Q22. Company registered hai but systematically wrong MRP print karti hai — software kaise pakdega?

Honest: pure software 100% nahi pakad sakta. Hum complaint-clustering + statistical anomaly ("40% scans me MRP disputes") + risk-score raise karke priority queue me target karte hain. Software suspect karta hai, human inspects.

💀 LEVEL 5 — KILLERIONS (Project Todne Ke Liye)

Q23. High-res printers + AI perfect typography copy kar lete hain — tab kya?



Pixels se pixels se nahi ladte. Defense-in-depth: (1) Print forensics — micro-text/guilloche patterns jo consumer printer render nahi kar sakta; (2) Cryptographic batch identity — digitally signed QR, cloned QR ka scan-velocity anomaly network-wide dikhta hai; (3) Behavioral intelligence — geography/velocity/complaint clustering. Detail Slide deck Section "Counterfeit Detection" me.

Q24. Customer website/kisi bhi site pe product details check kar sakta hai — tumhara overlap kya hai?

Website seller ki self-declaration hai — fraud tabhi hai jab PHYSICAL LABEL usse match na kare. Website apna data apne se compare karta hai (circular). Hum triple-source cross-verify karte hain: physical label (ground truth) × listing × registry. Mismatch = fraud flag.

Q25. Ek shelf me packets — one-by-one scanning realistic nahi.

Video-mode shelf sweep + perceptual-hash dedup (same frame skip) + same-batch consistency check (ek hi SKU ke MRPs compare, outlier flag). Risk-based sampling: high-risk brands 100%, others random. 30 min manual → ~3 min.

Q26. Fake QR/barcode jo dekhne me genuine lagta hai?

Trust nothing: GS1 checksum verify → registry lookup (company prefix registered?) → QR-encoded content vs printed label match → historical scan anomalies (same code 500× locations pe). Layered failure = layered defense.

Q27. Google/Microsoft ye kal launch kar dein to?

Big tech generic vision APIs banati hai; hamari value codified Indian law + enforcement workflow (notices, dashboards, risk queues) + violation-network effect me hai. Jaise CA software survive karte hain despite Excel — domain depth moat hai.

Q28. Agar tumhara system galat COTach charge kare aur media me case ban jaye?

Accountability design pehle se: human approval loop, dispute mechanism, immutable evidence, audit trails, insurance/waiver clause in govt contract, KPI-tracked FP rate. Hum decision-support tool hain, judge nahi.

Q29. 36–48 hours hackathon me itna sab kaise banaoge? (team skills?)

Deliberately cut MVP: single-language OCR, rules 4–7 + 9(6), basic dashboard, PDF report, vendor self-audit flow. *(Apni team ke actual roles yahan add karo — ML/fullstack/domain split.)* ora vision phased roadmap hai.

Q30. Success 6 mahine me kaise measure hoga?

Concrete KPIs: scans/inspector/day (>100 vs ~30 manual), violations/100 scans, FP rate <10%, time-to-report (days→minutes), repeat-offender brands ka compliance improvement %.

PART 2: 🌀 Judge Cross-Questions — Complete List (Jo Tumne Abhi Tak Nahi Socha)

Judge hackathons me typically **5 categories** me sawal maarte hain: Technical, Legal/Domain, Business, Data/ML, aur "Killer Questions" jo project todne ke liye hote hain. Sab cover karta hoon — har ek ke saath ready-made answer.

📁 CATEGORY 1: LEGAL & DOMAIN QUESTIONS (Sabse Zyada Yahan Se Aayenge)

Q1. "Legal Metrology rules state-wise alag hote hain kya? Ek rule engine kaise kaam karega?"

Answer:

"LM Act 2009 central law hai, lekin enforcement State Legal Metrology Departments karti hain aur kuch states me minor variations/additions hoti hain (jaise kuch states ne specific commodity rules add kiye). Isliye hamare RULES table me versioning hai aur rule ko 'jurisdiction' tag diya ja sakta hai — rule engine scan ke GPS/state ke hisaab se applicable rules hi load karta hai. Rules DB admin-updateable hai, code me hardcode nahi."

Q2. "Inspector ka report legally valid kaise hoga? Kya AI-generated violation notice court me stand karega?"

⚠️ Ye tough question hai.

Answer:

"Hamara system VIOLATION DETECT karta hai, notice ISSUE nahi karta. Notice sirf draft hota hai — final approval human Controller ka hai (hamare flow me approval loop already hai). Evidence image + timestamp + GPS + inspector ID ke saath digitally signed PDF banata hai, jisse tamper-proof documentation milti hai — ye manual system se zyada legally defensible hai kyunki manual memos me disputes hoti hain ki inspector ne sahi padha ya nahi."

Key phrase: "AI assists, human decides."

Q3. "Rule 9(6) font size measurement me error margin kitna hai? 1mm vs 0.8mm kaise reliably measure karoge camera se?"

⚠️ Technical trap — camera perspective distortion real issue hai.

Answer:

"Bina calibration ye impossible hai — hum isliye reference-object calibration use karte hain: barcode ki known dimensions (EAN-13 standard module width 0.33mm) ya coin/card edge se pixel-per-mm scale derive karte hain. Phir bhi $\pm 10-15\%$ error margin rehta hai. Isliye hamara system borderline cases ko 'MANUAL VERIFY' flag karta hai, automatic FAIL nahi deta. Critical violations pe inspector physical ruler se confirm karta hai — system evidence capture karta hai, judgment assist karta hai."

Q4. "2021 amendment me consumer care details mandatory hue the, 2017 me e-commerce rules aaye — rules change hote rahenge to tumhara system purana ho jayega."

Answer:

"Rules hamare DB entity hain, code nahi. Rule amendment aate hi admin UI se new rule add/update hota hai with effective_from date — historical scans purane rules ke against evaluate rahe, naye scans naye rules se. Ye exactly wahi problem hai jo

manual system me hota hai (inspector outdated rulebook use karta hai), humara centralized update ise solve karta hai."

Q5. "MRP overcharging label compliance hai ya retail practice? Tumhara system usko kaise address karega?"

Answer:

"Label PE declared MRP check karna hamara core scope hai. Overcharging (billing > MRP) detect karne ke liye Phase-2 extension possible hai: billing invoice OCR + MRP comparison. But wo retail audit domain me aata hai. Hamara focus declaration accuracy hai — kyunki agar label ka MRP hi galat hai (dual sticker), to consumer already deceive ho chuka hai."

📁 CATEGORY 2: TECHNICAL QUESTIONS

Q6. "OCR ki accuracy kitni hogi? Curved bottles, shiny wrappers, crumpled labels pe kya hoga?"

Answer:

"Flat labels pe PaddleOCR ~95%+ deta hai; curved/shiny surfaces challenge hai. Mitigations: (1) image quality gate pehle reject karta hai jo process hi na ho sake, (2) curved text ke liye CRAFT-style text detection + polar unwarping preprocessing, (3) confidence per field — low-confidence fields 'human verify' flag h hain, silently pass nahi hote. Honest expectation: ~90% end-to-end field extraction accuracy realistic hai, baaki human-in-the-loop."

Q7. "Offline mode meya-kya kaam karega? Poor connectivity wale warehouses me?"

Answer:

"On-device lightweight model (quantized MobileNet-based detector + small OCR) quick preview deta hai. Full results queue me store hote hain aur sync hone server-side validation complete hoti hai. Conflict case: offline preview vs result differ → server result authoritative."

Q8. "? Agar poore India me deploy ho to daily kitne scans honge? Cost?"

Answer:

"Est: 50,000 inspectors × 100 scans/day = 5M × 2MB 10TB/month). MeghRaj/NIC pe feasible; per-scan inference cost paise-level me aata quantized models se. Hackathon demo ke liye single sufficient."

Q9. "Duplicate product detection kaise? Same product different angle/lighting se photo li to system same SKU pechanega?"

Answer:

"Barcode visible ho to primary key. Nahi to perceptual hash + brand-logo embedding (CL-style similarity) + extracted-text fuzzy match ka composite fingerprint. Confidence se auto-link ya 'possible duplicate' review flag."

###10. "Model training data kahan se laoge? Annotated dataset ready-made available hai?"

⚠ Bahut common question.

:

"Ready-made LM(PC)-specific dataset exist NAHIarta — ye hi hamara moat hai. Approach: (1) synthetic data generation — template labels generate karke augmentations (blur, glare, rotation), (2) public datasets — COCO-text, IC15, Indian receipt datasets for OCR pretraining, (3) cold pe rules-based checks (presence/format) jo annotation-independent h, (4) production me inspectors' corrections se continuous improvement.-size ground truth ke liye printed test sheets at known sizes se calibration dataset banaya."

📁 CATEGORY 3: / IMPACT QUESTIONS

Q11. "Government ko is liye convince kaise karoge? Procurement cycle 2-3 saal leti hai."

Answer:

"Strategy: bottom-up pilot. Ek district me 10–20 inspectors ke saath 3-month pilot, before-after metrics (scans/day, violations detected revenue from penalties). Legal Metrology departments already digitization push hain. Alternative adoption path: e-commerce platforms aur FMCG brands 2B self-audit tool ke roop me adopt kare — wo faster buyers hain, government traction parallel build hota hai"

Q12. "Revenue model kya hai? Free rahega ya paid?"

Answer:

*"- Government: SaaS license/per-inspector seat or grant-funded

- Manufacturers: freemium self-audit (5 free scans/month → subscription) - E-commerce: bulk API pricing
- Consumer app (3): ads-free, possibly brand-sponsored verification badges
Penalty justification: states crores ka fine collect karte hain — se detection 100x badhega."

Q13 "Agar manufacturers apna data (compliance score) private rakhna chahein? DPDP concerns?"

Answer:

"Vendor apna data sirf khud dekhta hai — RBAC enforced. Government ko sirf inspection-derived violations dikhte hain (public enforcement matter hai). Personal data collect nahi hota (consumer PII). Audit logs immutable hain. DPDP Act design by default."

Q14. "Competitors? Art Flow jaisi companies ye kyun nahi bana r? Tumhare band kyun nahi lagega?"

Answer:

"Wo pre-print artwork tools hain — enterprise design teams ke liye, source-file based, ₹lakhs/year. Hum-market, field-deployable, rule-mapped-to-L-PC-2011, government-first system hain. Un paas enforcement workflow, GPS tagging, notice drafting, risk-scoringahi hai. Moat = codified Indian rule engine + violation network effect + government relationships. Aur haan, wo bhi b sakte hain — lekin hamara first-mover advantage in govt-specific workflows me hai."

📁 CATEGORY 4: ML/DATA QUESTIONS

Q15. "Font size measurement ka ground truth kaise validate karoge? Apne system ki kaise prove karoge?"

Answer:

"Controlled experiment: known font heights (1mm, 2mm, mm...) ke printed test sheets, different cameras/distances/lighting photograph karke error curve plot karenge. Target: ± 15 error at 90% confidence. Hackathon demo me ye calibration sheet live dikhayenge — judges ko measurable proof milega."

Q16. "False positives ka impact? Galat violation flag hua manufacturer ka business loss — accountability kaun lega?"

Answer:

"Three-tier output: CONFIRMED (high, multiple checks passed) / SUSPECTED (manual verify needed) PASS. Sirf CONFIRMED violations automated workflow me jaateain. Har violation ke saath raw evidence attached hai — manufacturer dispute kar sakta hai, dispute resolution flow hai. False positive rate track karenge as a KPI."

Q17. "Multilingual — Hindi, Tamil, Telugu sab handle karoge Regional languages me declarations bahut common hain."

Answer:

"PaddleOCR multilingual models support karta hai (H, Tamil, Telugu included). Hackathon me English+H, roadmap me top-6 regional languages. Rule engine

language-agnostic — wo extracted VALUES validate karta hai, text language se farak nahi padta (numbers universal hote hain)."

📁 CATEGORY 5: KILLER QUESTIONS (Project Todne Ke Liye)

Q18. 🗣️ "Agar ne label bilkul sahi print kiya but packet me kam bhara hai — tumhara system ye pakad lega? phir consumer protection me tumhara contribution limited nahi?"

Answer:

"Haan, actual fill-weight labcale ka domain hai. Lekin do cheezein: (1) statistical net-quantity checking via inspector spot-weighs can integrate as a data-entry field — system cross-references declared vs measured across batches flags systematic short-fill patterns, (2) hamara deliberately focused hai — ek system jo sab kuch kare, kuch bhi achha nahi karta. Declaration fraud India ka most issue hai aur wo unsolved hai. Depth over breadth."

###19. 🗣️ "Tumhara system rely karta photos pe. Inspector khud corrupt ho sakta hai — photo manipulate karke fake violation create kare, ya real violation hide kare?"

⚠️ Ye brilliant adversarial question hai.

Answer:

"Anti-tampering measures: (1) photo metadata (, GPS, device ID) cryptographically signed at capture on-device — EXIF editing detectable, (2) liveness check — photo camera stream se lena zaroori, gallery upload inspectors ke li disabled, (3) anomaly detection — ek inspector ka unusually/low violation rate flag hota hai, (4) audit log immutable hai. Perfect corruption-proof nahi, lekin manipulation cost dramatically badhta hai aur pattern-detectable ban jata hai. Manual system me ye accountability tha."

Q20. 🗣️ "Demo tum kya dikhao ge? Live internet, real products, OCR — ya sab mockup hai?"

Answer: (Ye prepare hona CRITICAL hai)

Demo plan suggest karo> 1. Real product packets (chips, shampoo sach, medicine box) live scan

2. Ek intentionally-viol sample (font chhota, missing date) prepared rakho — system usse pakde

3. Dual-MRP waala sample — tamper detection live

4. PDF report generation live

5. Offline mode dikhane ke liye airplane-mode trick

6. Backup: recorded video demo agar live fail ho jaye ⚠️

Q21. 🗣️ "3 mahine me ye sab build kar sakte ho? Team size kya hai, skills kya hain?"

Answer:

"Hackathon scope deliberately cut hai: single-language OCR, rules 4–9 subset, basic dashboard, PDF report. Poora vision phased roadmap hai. [Apni team ke skills bolo — e.g., 2 members CV/ML background, 1 fullstack, 1 domain research]."

Q22. 🗣️ "Consumer-facing appane ka plan hai? Warna consumer ko iska directayda kya hai?"

Answer:

"Indirect benefit pehle: better enforcement → cleaner market → consumer protected. Direct consumer feature Phase-3 me: scan-before-buy, complaint filing with auto-e (consumer ki photo hi complaint evidence ban jati hai — quality 10x improves). Currently complaints me consumers evidence provide n kar paate."

Q23. 🗣️ "Ag Google/Microsoft kal aisa product launch kar de to?"

Answer

"Big tech generic vision APIs banati hai — ham value codified Indian legal rules + enforcement workflow + government trust + violation-history network effect me hai. Google Lens label padh sakta hai, lekin Section 36 notice draft nahi kar sakta, repeat-off tracking nahi karta, state dashboards nahieta. Domain depth hamara moat hai — waise hi jaise GST filing software survive karte hain despite Google Sheets existing."#### Q24. 🗣️ "Success kaise measure karoge 6 months me?"

Answer:

Concrete KPIs:

- Scans per inspector per day (target: 100+, vs manual ~30)
 - Detection rate: violations found per 100 scans
 - False positive rate (<10%)
 - Time-to-reportmanual: days → ours: minutes)
 - Repeatender conversion: flagged brands me compliance improvement %

PART 3: Teen Sharp Questions — Detailed Answers

Ye teeno questions **judges bhi poochh sakte hain** — inke solid answers ready rakho. Chalo ek-ek karke:

? Q1: "Client website pe jaakar directly verify kar sakta hai, to tumhara OCR project overlap kaise hoga?"

Pehle samjho — website verification ka fundamental problem:

Website pe sirf WO information hoti hai jo SELLER khud daalta hai. Ye circular reasoning hai:

Seller ne listing me likha: "MRP ₹99, Net Qty 500g"



Tum website se verify karte ho



Website bolti hai: "MRP ₹99, Net Qty 500g" ?



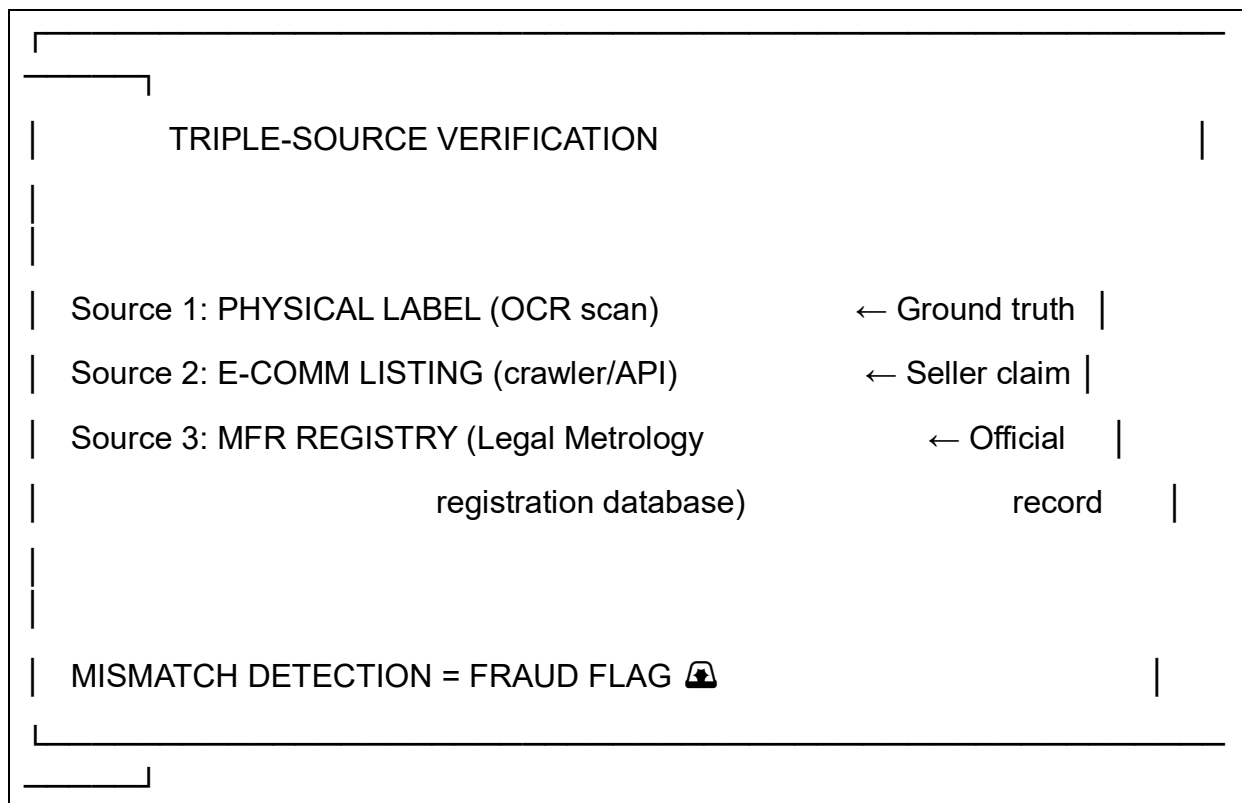
LEKIN PHYSICAL PACKET PE LIKHA HAI: "MRP ₹150, Net Qty 250g" ✗

Website = seller ki self-declaration. Physical packet = asli evidence. Fraud wahi hota hai jahan ye dono match NAHI karte!

Real Evidence (ye judges ko quote karo):

1. **CCPA ne 2021-23 me Amazon, Flipkart, Meesho etc. ko notices diye** kyunki e-commerce listings pe declarations missing/galat the — matlab platforms ke apne systems hi violations catch nahi kar paate.
2. **LM (PC) Amendment Rules, 2017 & 2022** specifically e-commerce display rules banane pade — government ne maana ki self-regulation fail ho rahi hai.
3. E-commerce pe **lakhs of sellers** hain; platform har listing manually verify nahi kar sakta.

Hamara system website se OVERLAP nahi, CROSS-VERIFY karta hai 🖱️



Example: Listing bolta hai MRP ₹99 → packet photo pe ₹149 dikhta hai → mismatch flag + evidence image ke saath report. Website akela ye KABHI detect nahi kar sakta kyunki uske paas physical packet nahi hai.

One-liner for judges:

"Website verification seller ke apne data ko seller ke data se compare karta hai — fraud detect karne ke liye independent source chahiye. Humara ground truth physical label hai."

🔗 Q2: "Dual price / bulk purchase — 10 packets one-by-one check karna time-consuming, na karo to fraud. Kaise solve?"

Pehle clarify: Dual-MRP detection ka asli matlab

Dual-MRP problem mostly **ek hi packet pe do stickers** (old MRP sticker ke upar new MRP sticker) ya **same batch me different MRPs** hota hai. Har customer 10 packets individually check nahi karega — sahi point hai.

Solutions (layered approach):

◆ **Layer 1: Batch Scan Mode (Inspector ke liye)**

CONTINUOUS BATCH CAPTURE MODE	
Inspector shelf ke saamne khada	
App camera ON → video mode	
Frame 1 →	auto-detect packet → scan
Frame 2 →	auto-detect packet → scan
... (auto-deduplication via perceptual hashing)	
10 packets = ~60 seconds total	
(5-6 sec per packet, parallel)	

- Video stream se frames nikalte hain, CV model automatically packet labels detect karta hai
- **Perceptual hash (pHash)** se duplicate frames skip hote hain
- Ek shelf sweep = poore shelf ka compliance status
- Manual inspection: ~30 min/shelf → Hamara: ~3 min/shelf (**10x faster**)

◆ Layer 2: Batch-Level Consistency Check (Smart trick)

Ek hi SKU/batch ke multiple scans ko **group** karo aur compare karo:

Same barcode + same batch code detected:

Packet 1 → MRP ₹99 📄

Packet 2 → MRP ₹99 📄

Packet 3 → MRP ₹120 🚨 MISMATCH!

→ Auto-flag: "Batch inconsistency detected"

→ Evidence: dono annotated images side-by-side

Ye **statistical outlier detection** hai — ek packet galat MRP ka ho to group me turant pakda jata hai.

◆ Layer 3: Sticker-Tamper Detection (Single packet pe)

CV-based anomaly detection:

- **Edge detection:** Do overlapping stickers = double edges/rectangular seams around price area
- **Color/texture discontinuity:** Naye purane sticker ka paper tone alag hota hai
- **Font mismatch:** Sticker ka font vs original printed font alag
- **Sticker boundary detection:** YOLO model train on tampered-sticker dataset

◆ Layer 4: Consumer-side spot check (Future scope)

QR scan by consumer → app instantly shows "This product's last 100 scans showed MRP ₹99 — you're seeing ₹120? Report."

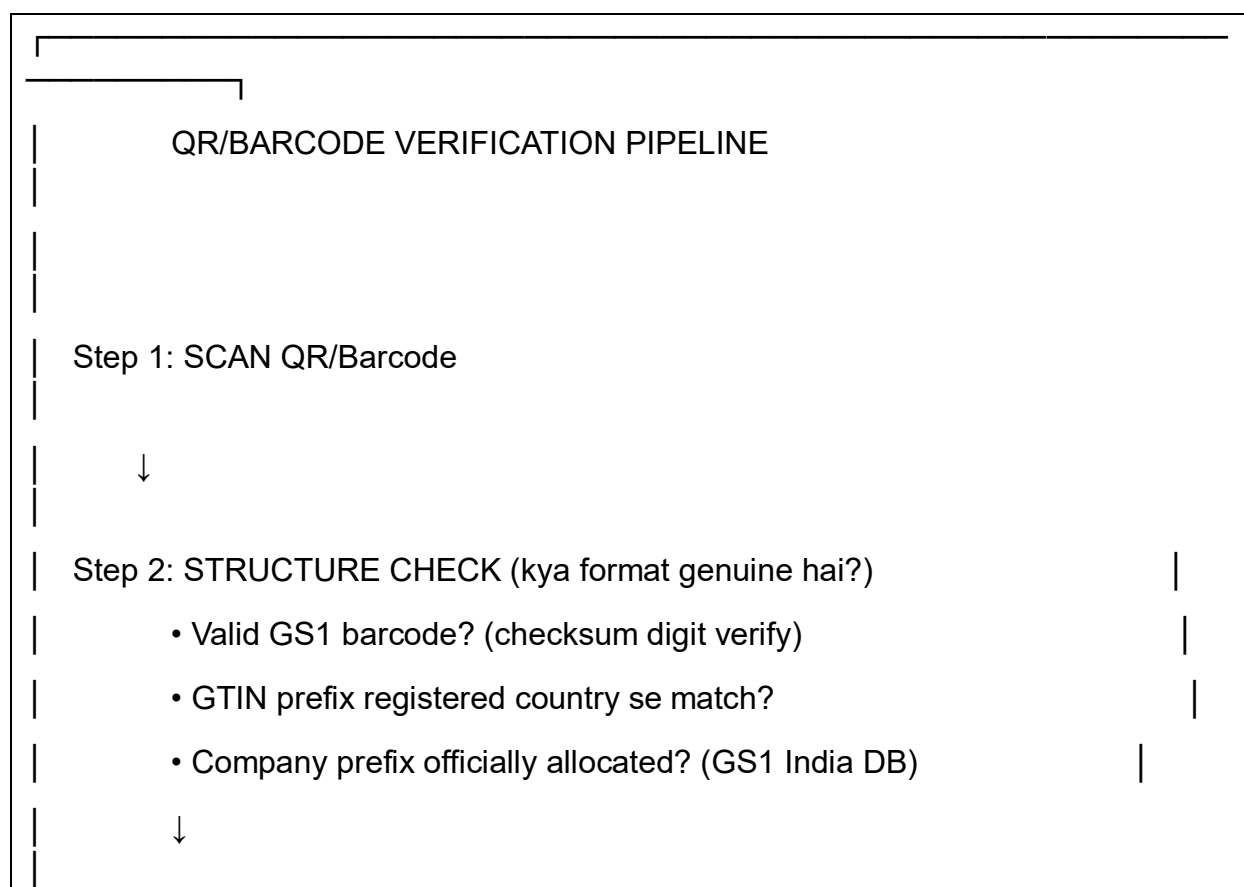
Time math honest rakho:

"Har packet individually full-check nahi karna padta. Batch mode me shelf-sweep 3 min leta hai, aur statistical consistency check se ek hi batch ke andar anomalies pakde jate hain. Random sampling + risk-targeting (high-risk brands pehle) standard enforcement practice hai — hum usse automate karte hain."

🔗 Q3: "Company QR/barcode me galat info daal de jo SAHI dikhta hai — kaise verify hoga?"

Ye sabse achha question hai. Iska answer **trust hierarchy** hai:

Core Principle: QR/barcode content ko TRUST mat karo — VERIFY karo against independent sources



	Step 3: REGISTRY CROSS-CHECK	
	• Barcode → Legal Metrology manufacturer registry • Kya ye company registered hai? LUT number valid? • Barcode company ke naam pe issued hai ya kisi aur ke?	
	↓	
	Step 4: CONTENT SANITY CHECK	
	• QR me encoded data vs PHYSICAL label data match? • MRP format valid? Date logical? (expiry > mfg date) • Net quantity unit standards se match?	
	↓	
	Step 5: HISTORY CHECK	
	• Is barcode ke past scans ka record? • Achanak MRP/date change hua? (relabeling pattern)	
	↓	
	RESULT: ☒ Verified ☐ Suspicious ☐ Fraud Flagged	

Concrete fraud examples jo pakde jayenge:

Fraud Attempt	Detection Mechanism
Fake barcode (kisi aur company ka copy)	GS1 checksum + company prefix mismatch
Barcode registered nahi	Registry lookup fail → flag
Registered company ne galat MRP encode kiya	Physical label vs QR data mismatch + registry MRP history

Fraud Attempt	Detection Mechanism
Expired product ka date change karke relabel	Month-year logic check + batch history ("ye batch 2023 ka tha, ab 2025 likha hai?")
QR me link fake website pe le jata hai	URL domain validation against official company domains

⚠️ Honest Limitation (judges ko khud batao — trust badhega):

"Agar manufacturer Level-1 se hi fully fraudulent hai — registered company, valid barcode, but wo jaan-boojh kar galat MRP print kar raha hai — to pure software se 100% detection possible nahi hai. Us case me hamara system:

1. Consumer complaints ko us brand se LINK karta hai (complaint clustering)
2. Statistical anomaly dikhata hai ('is brand ke 40% scans me MRP disputes')
3. Risk-score raise karke priority inspection queue me daal deta hai

Matlab software FRAUD PROVE nahi karta — fraud SUSPECT karke human inspector ko targeted evidence deta hai. Ye hi realistic aur legally sound approach hai."

🔗 Judges Ke Liye Summary One-Liners :

Question	Killer Answer
Website se verify?	"Website = seller ka apna data. Fraud tabhi hai jab physical label alag bole. Hum triple-source cross-verification karte hain — physical label ground truth hai."

Question	Killer Answer
10 packets check?	<i>"Video-mode shelf sweep + perceptual hash dedup + same-batch consistency check. 30 min ka manual work 3 min me, aur batch-level outliers automatic pakde jate hain."</i>
Fake QR/barcode?	<i>"Trust nothing — verify everything: GS1 checksum → registry lookup → QR-vs-label content match → historical anomaly detection. Aur jo hum nahi pakad sakte, use risk-scoring se inspector tak target karte hain."</i>
